

# JEE MAIN REVISION MASTER: SURFACE CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                                     | Frequency | Category            |
|---------------|------------------------------------------------------------------|-----------|---------------------|
| High          | Adsorption: Physisorption vs Chemisorption & Freundlich Isotherm | 11 PYQs   | Surface Chemistry   |
| High          | Colloids, Tyndall Effect & Hardy-Schulze Rule                    | 8 PYQs    | Colloidal Chemistry |

## 2. Master Formula Reference Card

| Concept / Topic                                                  | Mathematical Formulation & Governing Equations                                                                                                      |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Adsorption: Physisorption vs Chemisorption & Freundlich Isotherm | $\frac{x}{m} = k P^{1/n}$ , $\log \frac{x}{m} = \log k + \frac{1}{n} \log P$<br>(Slope = $1/n$ )                                                    |
| Colloids, Tyndall Effect & Hardy-Schulze Rule                    | Coagulating Power of Cations: $\text{Al}^{3+} > \text{Ba}^{2+} > \text{Na}^{+}$ ,<br>Coagulating Value $\propto \frac{1}{\text{Coagulating Power}}$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Adsorption: Physisorption vs Chemisorption & Freundlich Isotherm (High Yield — 11 PYQs)

**Core Principle:** Physisorption involves weak van der Waals forces, is reversible, multi-layered, non-specific, and decreases continuously with temperature. Chemisorption involves chemical bond formation, is irreversible, unilayered, highly specific, and initially increases then decreases with temperature. Freundlich isotherm:  $(x / m) = k P^{1/n}$  ( $0 \leq 1/n \leq 1$ ).

■ **Caution / JEE Pitfall:** Chemisorption has high enthalpy of adsorption (80-240 kJ/mol), whereas physisorption has low enthalpy (20-40 kJ/mol).

■ **Speed Trick:** Easily liquefiable gases (high critical temperature  $T_c$ , e.g.  $\text{SO}_2$ ,  $\text{NH}_3$ ) are adsorbed in greater amounts by charcoal than permanent gases ( $\text{H}_2$ ,  $\text{N}_2$ ).

#### Representative Source PYQs:

- [2023 Shift] Q#1 (MCQ): "For Freundlich adsorption isotherm, a graph of  $\log(x/m)$  vs  $\log(p)$  gives a straight line. The slope of line and its Y-axis intercept r..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "Aqueous solution of raw sugar when passed over beds of animal charcoal, it becomes colourless. Pick the correct set of terminologies that ca..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "Which can adsorb larger volume of hydrogen gas?..." [Ans: N/A]

### 3.2 Colloids, Tyndall Effect & Hardy-Schulze Rule (High Yield — 8 PYQs)

**Core Principle:** Colloid classification: Lyophilic (solvent-loving, reversible, self-stabilizing, e.g. starch, gelatin) vs Lyophobic (solvent-hating, irreversible, e.g. metal sols, As<sub>2</sub>S<sub>3</sub>). Properties: Tyndall effect (scattering of light), Brownian motion (kinetic motion preventing settling), Electrophoresis (confirms charge on colloidal particles). Hardy-Schulze Rule: Coagulating power of an electrolyte is directly proportional to the 4th power of the valency of the active ion ( $Z^4$ ).

■ **Caution / JEE Pitfall:** Gold number is INVERSELY proportional to protective power: a protective colloid with a SMALLER gold number has HIGHER protective efficiency (Gelatin has lowest gold number implies best protector).

■ **Speed Trick:** For negatively charged As<sub>2</sub>S<sub>3</sub> sol, Al<sup>3+</sup> has the greatest coagulating power; for positively charged Fe(OH)<sub>3</sub> sol, [Fe(CN)<sub>6</sub>]<sup>4-</sup> has the greatest coagulating power.

**Representative Source PYQs:**

- [2024 Shift] Q#1 (MCQ): "Identify the incorrect statement..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "For the coagulations of positively charged hydrated ferric - oxide sol, the flocculating power of the ions is in the order:..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Gold sol is not a..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2024 (Q#1)

Identify the incorrect statement

- (A) Values of colligative properties of colloidal solution are of small order compared to values of true solution.
- (B) Tyndall effect is observed only when diameter of the dispersed particles is not much smaller than wavelength of incident light.
- (C) Colour of colloidal solution depends on the wavelength of light scattered by the dispersed particles.
- (D) Brownian movement is due to balanced bombardment of molecules of dispersion medium on colloidal particles.

**Correct Answer: None**

**Solution Analysis:** Among the given statements incorrect statement is given in option (d). Its correct form is "Brownian movement is due to unbalanced bombardment of molecules of dispersion medium on colloidal particles".

### Benchmark Problem 2: KCET 2023 (Q#1)

For Freundlich adsorption isotherm, a graph of  $\log(x/m)$  vs  $\log(p)$  gives a straight line. The slope of line and its Y -axis intercept respectively are

- (A)  $\log((1/n)^k)$
- (B)  $(1/n), \log k$
- (C)  $\log((1/n)), \log k$
- (D)  $(1/n), k$

**Correct Answer: None**

**Solution Analysis:** We know that, Freundlich adsorption isotherm equation is given by  $\log(x/m) = \log k + (1/n) \log p$  ..... (i)  
Also equation of straight line is  $y = mx + c$  ..... (ii) On comparing Eq. (i) and Eq. (ii), we get Slope,  $m = (1/n)$  Intercept,  $c = \log k$